Cryptography

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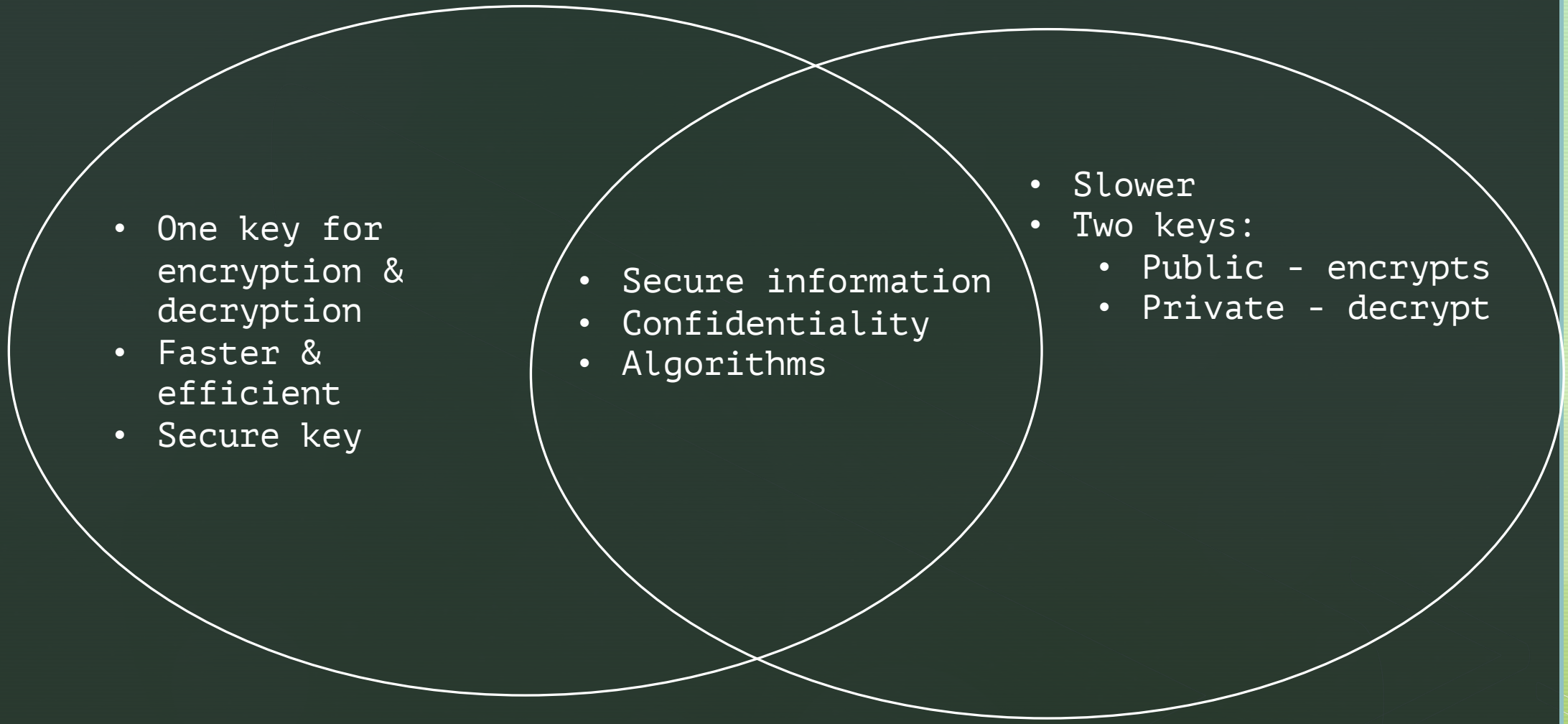
What is Cryptography?

- The practice of securing information by transforming it into an unreadable format
- Ensures that only authorized users can access and interpret it
- Encryption: $C = E(m, k)$
- Decryption: $m = D(C, k)$

Core Principles

- Confidentiality
  - Ensures that information is only accessible to authorized users
- Integrity
  - Ensures that information is accurate and unaltered throughout its cycle
  - Unauthorized changes can be detected
- Authentication
  - Verifies the identity of a user or system is valid
- Non-repudiation
  - Proof or origin and delivery
  - Ensures a party from denying their previous actions
  - Sender cannot deny signing or sending a message

Symmetric Vs. Asymmetric



Mathematics

- Mathematics in cryptography show up in serval branches:
  - Number theory
  - Algebra & Linear Algebra
  - Probability & Statistics

Caesar Cipher

- Technique used by Julis Caesar
  - Works by shifting the letter in the plaintext message by a certain number of positions
    - Shift or key
- How to solve:
  - Find the key
  - Shift letters back
  - Decode each letter

Hill Cipher

- Technique is based on linear algebra
- Each letter of the alphabet is represented by a number
  - A = 0, B = 1, … , Z = 26
- Encryption:
  - Divide plaintext into blocks of (n) letters
  - Convert letters to numbers
  - Each block needs to be written as a column vector
  - Use a key matrix (n X n)
  - Multiply key matrix by each vector
  - Take modulo 26
  - Convert numbers back to letters

References

- https://www.geeksforgeeks.org/computer-networks/cryptography-introduction/
- https://www.understandthemath.com/blog/math-behind-cryptography
- https://www.geeksforgeeks.org/dsa/hill-cipher/